

L4.2 Adversarial Attacks for AD Quiz

1. When the victim is a full driving stack instead of an image classifier, what replaces the small ℓ_p -norm ball as the perturbation constraint?

- A) A smaller learning rate for the attacker's optimizer, so the attack stays stealthy
- B) A requirement that the perturbation is invisible to every camera sensor
- C) A limit on how many gradient steps the attacker may run per frame
- D) Physical realizability: the attack must be a printable, projectable or placeable object

ANS:

2. Why is fooling an object detector in a single frame usually not enough to change what the planner sees?

- A) Planners ignore camera detections and rely on HD maps to find all objects
- B) The tracker is a temporal low-pass filter: isolated misses are bridged by prediction
- C) Detectors are re-trained online after every frame that was attacked successfully
- D) Non-maximum suppression removes every adversarial box before the tracking step

ANS:

3. In the lecture's three levels of attack success, what is a Level 2 (persistent perception failure)?

- A) A single frame that is misclassified, measured by the attack success rate of the model
- B) A collision or hard braking event, measured by the collision rate of the vehicle
- C) An error that survives tracking and fusion over many frames (track loss, ID switches)
- D) A change of the model weights, measured by the accuracy drop on a held-out test set

ANS:

4. Which metric belongs to Level 3 (closed-loop safety violation)?

- A) Collision rate or time to violation of a safety rule
- B) The mAP drop of the detector on a held-out data set
- C) The number of ID switches of tracks within one video
- D) The attack success rate measured on a single frame

ANS:

5. Toy tracker (constant-velocity Kalman filter, 10 Hz, max_age = 5 frames). What happens if detections of an object are suppressed for 2 frames versus 8 frames?

- A) 2 frames: the track is deleted; 8 frames: no visible change to the downstream modules
- B) 2 frames: no visible change; 8 frames: the track is deleted and a new ID appears
- C) Both gaps delete the track immediately and the object is lost forever afterwards
- D) Neither gap has any effect, because a Kalman filter never forgets any track at all

ANS:

6. What did Jia et al. (ICLR 2020, 'Fooling Detection Alone is Not Enough') find about attacks on tracking?

- A) Trackers are more vulnerable than detectors, so 10 % detection success is enough
- B) Detection attacks often fail to change tracking; over 98 % detection success may be needed
- C) Tracking-by-detection cannot be attacked at all in the digital domain
- D) Hungarian association can be inverted exactly by an attacker with white-box access

ANS:

7. Why do Ma et al. (VehicleSec 2023, WIP) use dynamic adversarial patches against multi-object tracking?

- A) Dynamic patches are much cheaper to print than static patches are
- B) Dynamic patches need no knowledge of the perception pipeline at all
- C) Static patches are always detected and removed by the Kalman filter
- D) A static patch cannot adapt to the tracker's state from frame to frame

ANS:

8. Ma et al. report 90 % and 80 % success rates on 10 BDD video clips. How should these numbers be quoted?

- A) As proof that deployed vehicles crash in 80 to 90 % of their real encounters
- B) As the success rate of a printed physical patch measured on real public roads
- C) As preliminary, white-box, digital results, not as a general risk for deployed vehicles
- D) As the average success over all tracker designs that are used in the industry

ANS:

9. Why can a LiDAR not tell a real reflection from an injected one in a laser spoofing attack?

- A) Its processor cannot compute the time of flight accurately enough for this
- B) It only accepts returns whose intensity is higher than a fixed threshold
- C) It computes $R = c \cdot \Delta t / 2$ and trusts any return with the right wavelength and timing
- D) It cannot measure the angle under which a return arrives at the sensor

ANS:

10. Cao et al. (CCS 2019) showed that blind LiDAR spoofing is insufficient. What was the fix?

- A) Optimization with global sampling to find point patterns accepted as a vehicle
- B) Injecting many thousands of random points to saturate the object detector
- C) Physically blocking the laser beam with a large piece of black cardboard
- D) Replacing the weights of the victim's detector through an over-the-air update

ANS:

11. What physical invariant does the occlusion-pattern defense (CARLO, Sun et al., USENIX Security 2020) exploit?

- A) Real vehicles always reflect with a higher intensity than the ground points around them
- B) Real vehicles block the laser, so there are no returns behind them; spoofs leave it populated
- C) Real vehicles always produce exactly 70 points in a single LiDAR scan of the scene
- D) Spoofed points always arrive earlier than the returns of real objects in the scene

ANS:

12. Why does the removal attack of Cao et al. (USENIX Security 2023, 'You Can't See Me') defeat even a robustly trained detector?

- A) It adds an adversarial patch that the network is trained to misclassify
- B) It re-trains the detector on poisoned data during the next software update
- C) It deletes the target's returns at the sensor, so the network never sees the object
- D) It jams the GPS receiver so that detections cannot be localized

ANS:

13. Tu et al. (CVPR 2020) hide a vehicle from LiDAR detectors with which type of attack?

- A) A laser that injects phantom points in front of the victim car
- B) A universal 3D adversarial mesh placed on the vehicle's roof
- C) A fabricated V2X message that carries a fake list of objects
- D) A sticker placed on the lens of the front camera of the victim

ANS:

14. In the SenSys 2021 attack on LiDAR segmentation (Zhu et al.), what is the purpose of the occlusion loss L_{occ} ?

- A) To keep the features of the unrelated parts of the scene unchanged
- B) To maximize the distance between predicted and target labels of points
- C) To limit the adversarial object to at most ten points in total
- D) To keep the object from being blocked, so the laser can actually hit it

ANS:

15. What does the trajectory-prediction attack of Lou et al. (USENIX Security 2024) achieve?

- A) It hides every vehicle in the scene from the LiDAR object detector of the victim
- B) It disables the lateral controller of the victim vehicle during a lane change
- C) Cardboard near a parked car makes the predictor expect it to move, causing unsafe avoidance
- D) It deletes the HD map from the memory of the victim vehicle while driving

ANS:

16. What makes the frustum attack of Hallyburton et al. (USENIX Security 2022) effective against camera–LiDAR fusion?

- A) It spoofs points inside the camera frustum of an object, keeping both sensors consistent
- B) It spoofs points far outside any camera frustum, away from every real object nearby
- C) It blinds the camera with a bright light so that only the LiDAR input remains
- D) It needs white-box access to the weights of the camera-LiDAR fusion network

ANS:

17. How does MSF-ADV (Cao et al., IEEE S&P 2021) differ from the frustum attack?

- A) It attacks only the camera branch of the fusion system and never touches the LiDAR
- B) It spoofs points with a laser in exactly the same way as the frustum attack
- C) It attacks the V2X link between the vehicles rather than the on-board sensors
- D) One 3D-printed object is optimized against the camera and the LiDAR branch at once

ANS:

18. In the dirty road patch attack on lane centering (Sato et al., USENIX Security 2021), which stage is attacked and how is harm defined?

- A) The lane-detection DNN; harm is the lateral deviation of the vehicle
- B) The lateral controller; harm is the classification accuracy of the network
- C) The GPS module; harm is the localization drift of the vehicle
- D) The planner's cost map; harm is the route length driven by the vehicle

ANS:

19. Why is Acero (Song et al., USENIX Security 2023) not stopped by defenses such as adversarial training or sensor consistency checks?

- A) It uses only realistic, legal driving maneuvers; no pixel, point or signal is modified
- B) It only works against vehicles that do not carry any perception sensors at all
- C) It modifies the neural network weights of the vehicle through a fake software update
- D) It relies on a laser attack against the front camera of the victim vehicle

ANS:

20. Why do numbers such as CARLO's 5.5 % residual attack success overstate robustness?

- A) They were measured on a data set that does not contain any vehicles at all
- B) Defenses cannot be evaluated by the success rate of an attack in any setting
- C) A residual success rate that is below 10 % does not have any statistical meaning
- D) They come from the published, non-adaptive attack; an adaptive attacker can model the check

ANS: